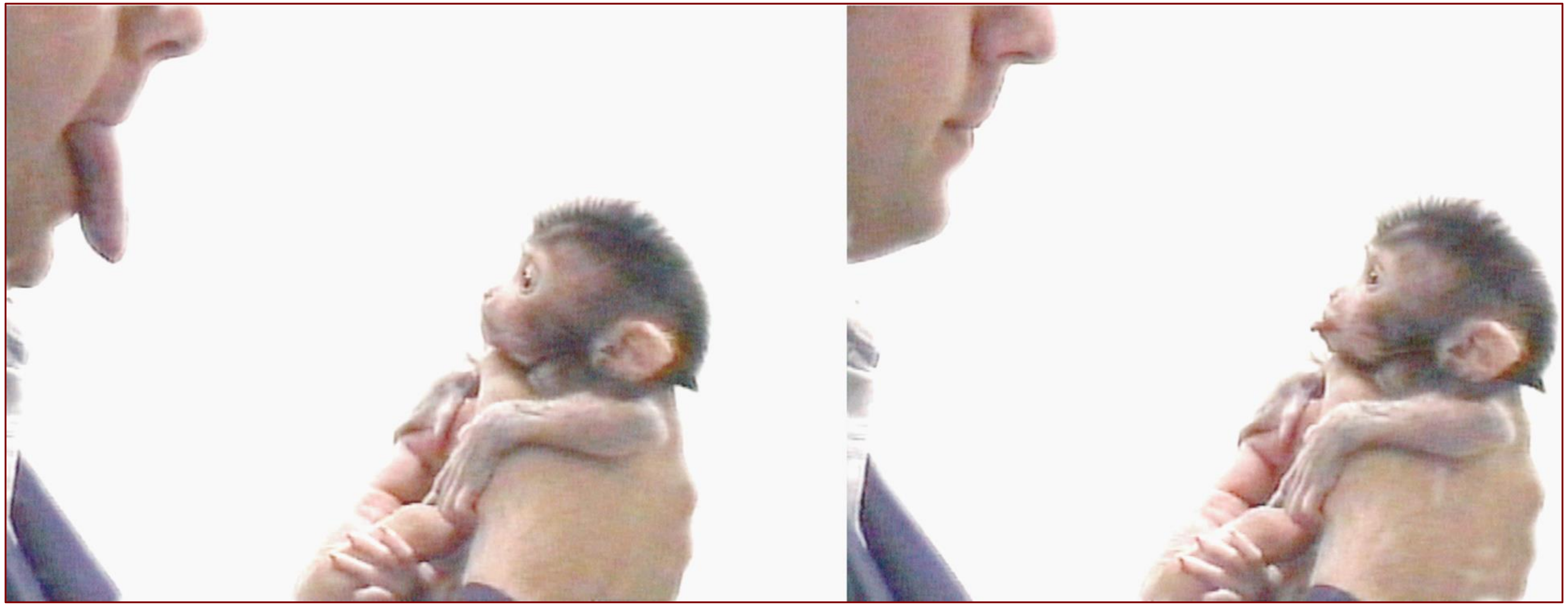
16-831: Introduction to Robot Learning, Fall 2024

Imitation Learning



https://en.wikipedia.org/wiki/Imitation

Instructor: Guanya Shi

Learning and Control for Agile Robotics Lab

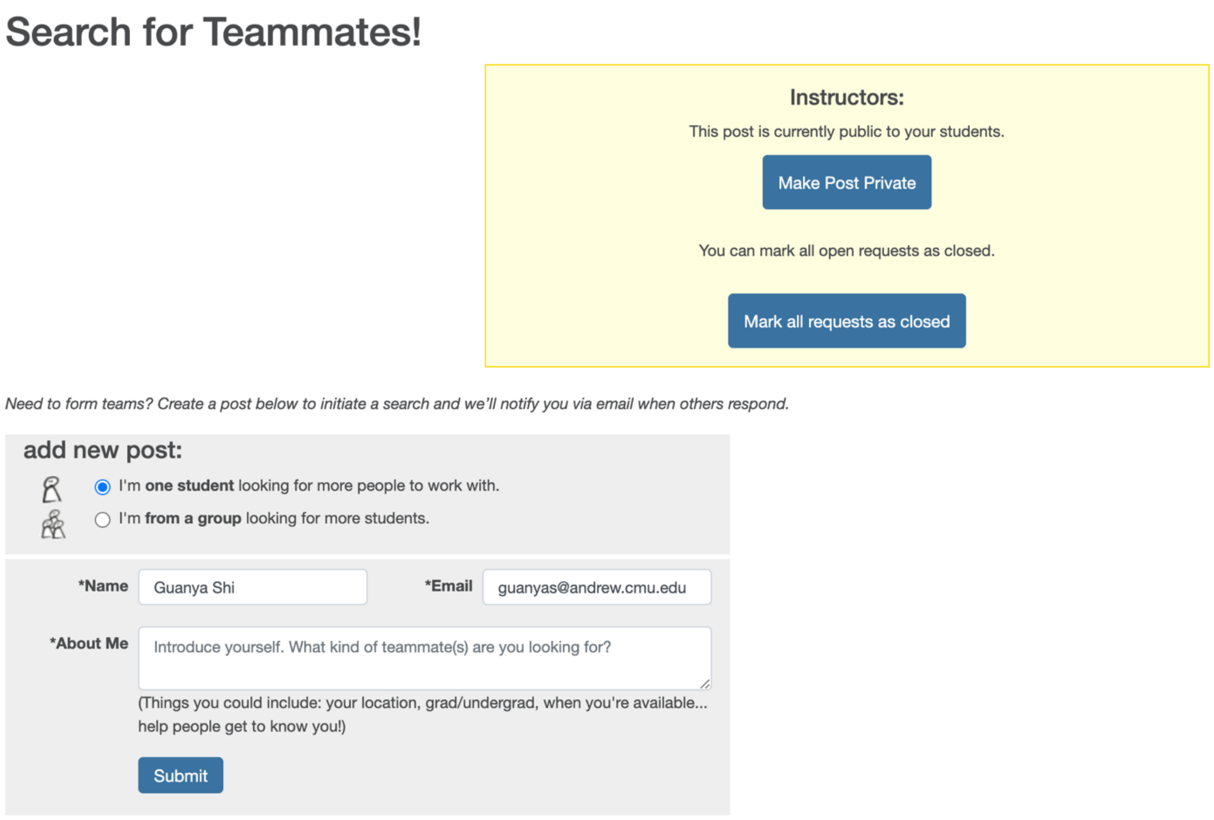
Carnegie Mellon University
The Robotics Institute

Recording Reminder

- This session will be audio recorded for educational use by other students in this course.

Project

- ❑ Please carefully read the Piazza post!
- ❑ Start thinking about your project and searching for teammates!
  - Project proposal deadline: Oct 11
- ❑ AWS credit request form: deadline this Friday (Sep 13)



Course Structure

Latest schedule: https://16-831.github.io/fall24/lectures

- ❑ Topic group 1: Course introduction: What is robot learning and why?
- ❑ Topic group 2: Machine/Deep learning refresher
- ❑ Topic group 3: Imitation learning
- ❑ Topic group 4: Model-free reinforcement learning
- ❑ Topic group 5: Model-based reinforcement learning
- ❑ Topic group 6: Bandit, preference-based learning, exploration
- ❑ Topic group 7: Reinforcement learning from offline data (one guest lecture)
- ❑ Topic group 8: Specialized topics (one guest lecture)
- ❑ Project presentation

Outline

- ❑ Notations, MDP, POMDP
- ❑ What is imitation learning?
- ❑ Why/how is imitation learning different from standard supervised learning?
  - Does behavior cloning (BC) work?
- ❑ How to address the issue?
- ❑ Deep imitation learning via generative models

(Some slides are from 22F, 23F, Berkeley CS 285, and CMU 10-703)

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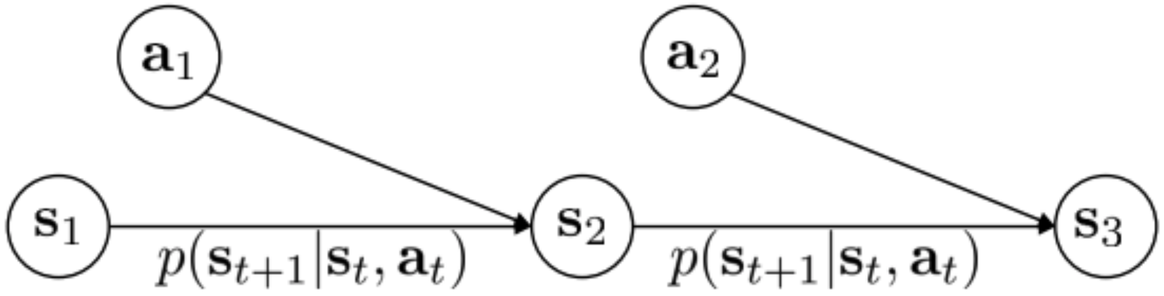
MDP

- Definitions:
  - S is the state space. $s_t \in S$ is the state at time step t.
  - A is the action space. $a_t \in A$ is the action at time step t.
  - O is the observation space. $o_t \in O$ is the observation at time step t.
  - p is the one-step transition probability (a.k.a., dynamics): $s_{t+1} \sim p(\cdot \mid s_t, a_t)$
  - h is the observation model: $o_t \sim h(\cdot \mid s_t)$
  - $r: S \times A \to \mathbb{R}$ is the reward function.

- A Markov decision process (MDP) is a tuple (S, A, p, r)
  - Goal: learn a policy $\pi_\theta(a_t \mid s_t)$



POMDP

- Definitions:
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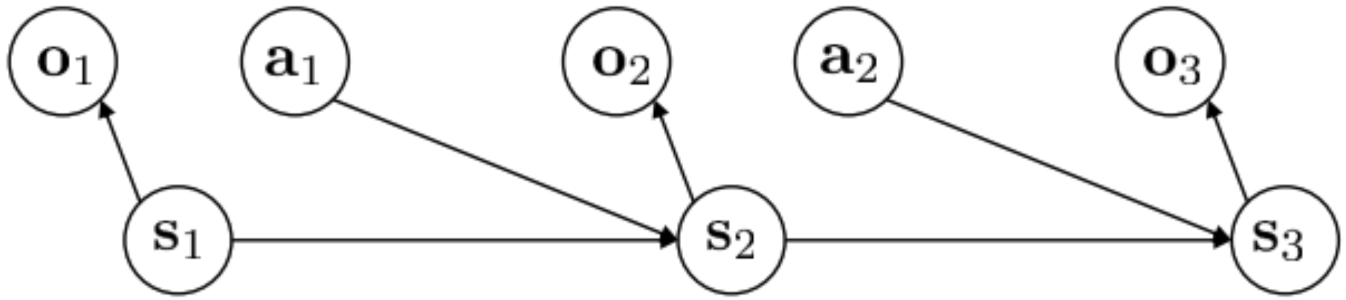
  - h is the observation model: $o_t \sim h(\cdot \mid s_t)$
  - $r: S \times A \to \mathbb{R}$ is the reward function.

- A partially observed Markov decision process (POMDP) is a tuple (S, A, O, p, h, r)
  - Goal: learn a policy $\pi_\theta(a_t \mid \text{observations})$



Markov Property

- Definitions:
  - S is the state space. $s_t \in S$ is the state at time step t.
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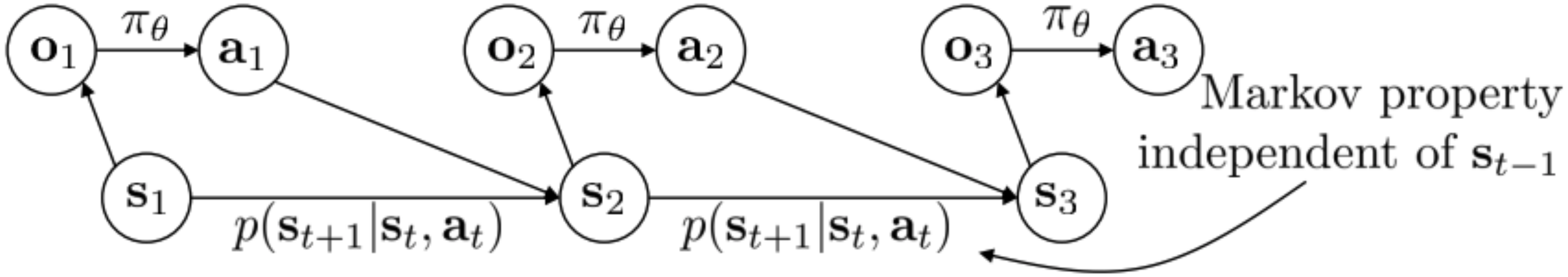
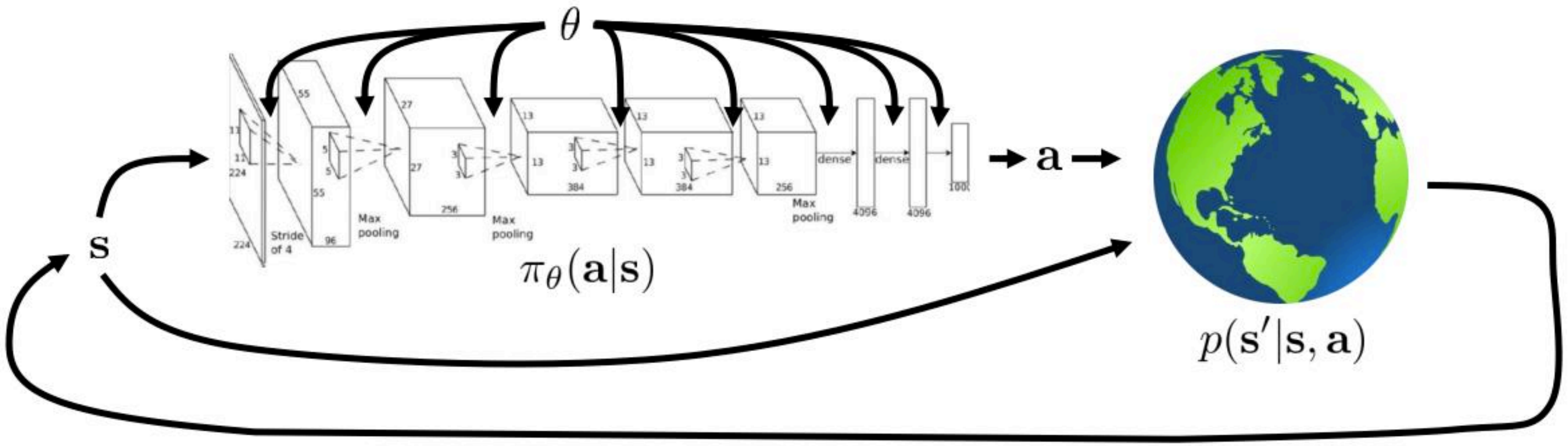
- Why Markov?
  - Allow us to throw away history once state is known!




Andrey Markov

The Goal of Reinforcement Learning and Control

- Finite horizon case: T is finite
- Infinite horizon case: $T = \infty$
- The cumulative reward can be discounted: $\sum_t \gamma^t r(s_t, a_t)$ where $0 < \gamma \leq 1$



$$\underbrace{p_\theta(\mathbf{s}_1, \mathbf{a}_1, \ldots, \mathbf{s}_T, \mathbf{a}_T)}_{\pi_\theta(\tau)} = p(\mathbf{s}_1) \prod_{t=1}^{T} \pi_\theta(\mathbf{a}_t|\mathbf{s}_t) p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t)$$

$$\theta^\star = \arg\max_\theta E_{\tau \sim p_\theta(\tau)} \left[\sum_t r(\mathbf{s}_t, \mathbf{a}_t) \right]$$

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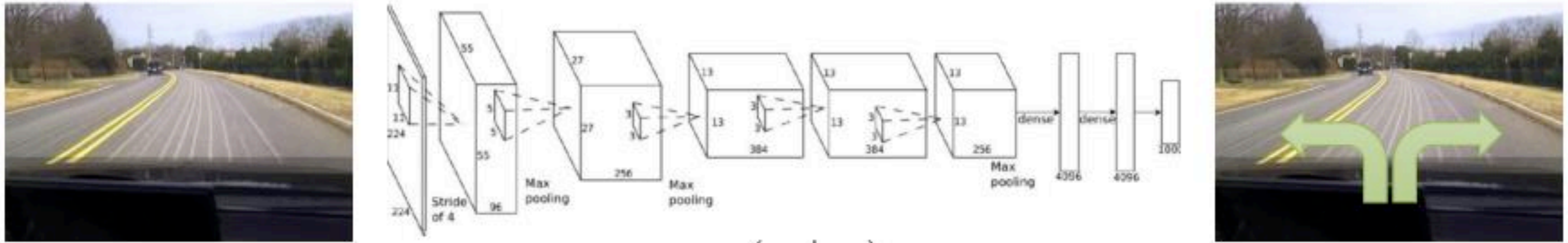
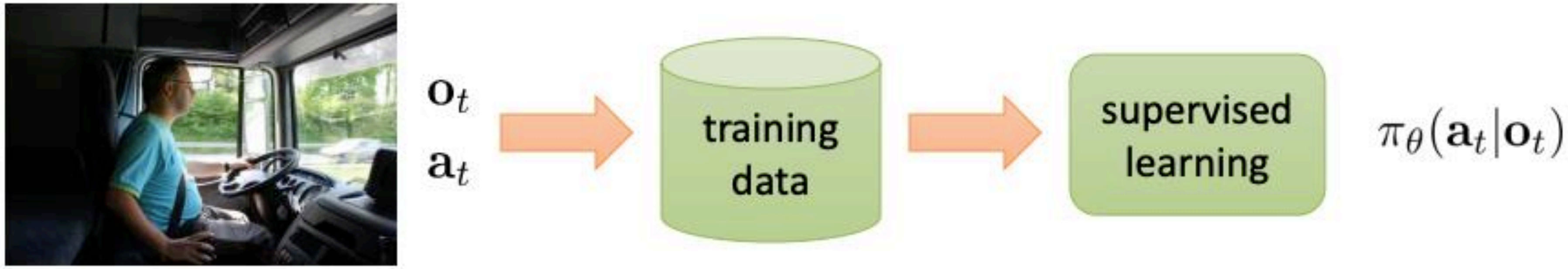
Imitation Learning

❑ Main idea:

- Collect expert data (observation/state and action pairs)
- Train a function to map observations/states to actions



$\mathbf{o}_t$ $\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$ $\mathbf{a}_t$



Deep Imitation Learning in 1989

- A CMU paper!
  - CMU has incubated many self-driving companies

ALVINN:
AN AUTONOMOUS LAND VEHICLE IN A NEURAL NETWORK

Dean A. Pomerleau
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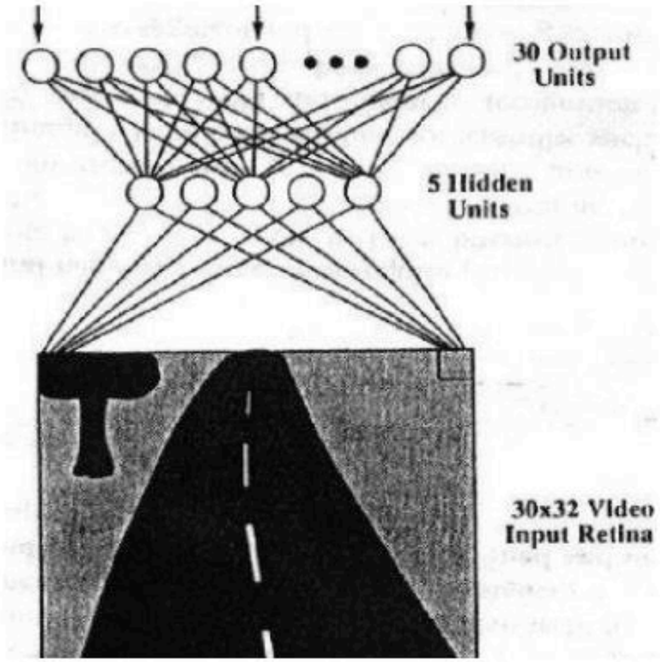
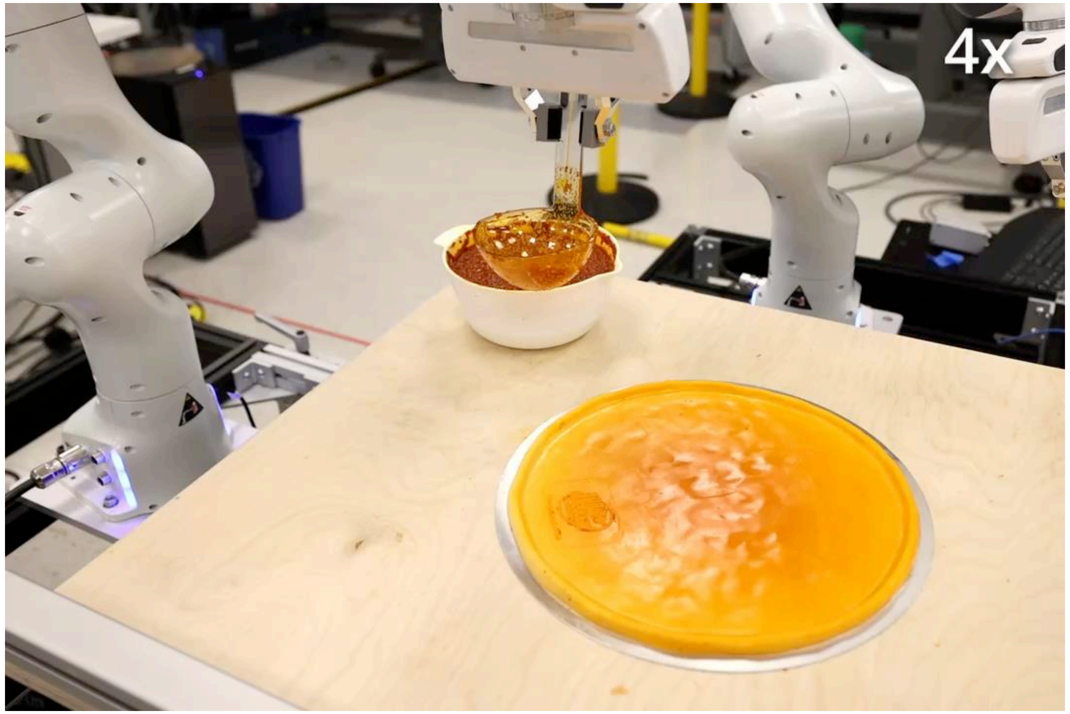


Deep Imitation Learning Today

- Left: Mobile ALOHA; Right: Diffusion policy.



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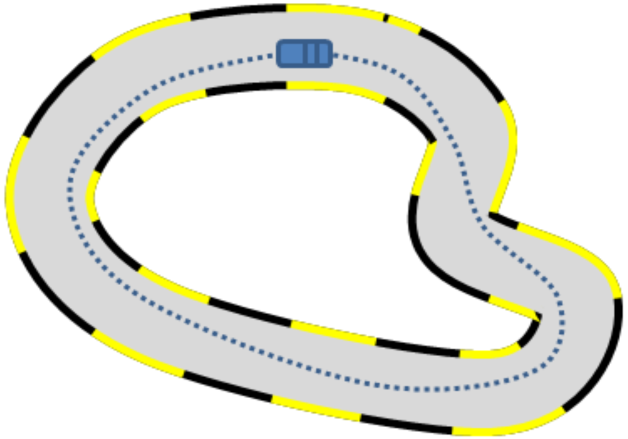
It Sounds Like Supervised Learning …

- Main idea:
  - Collect expert data (observation/state and action pairs)
  - Train a function to map observations/states to actions
  - A.k.a. “**behavior cloning (BC)**”

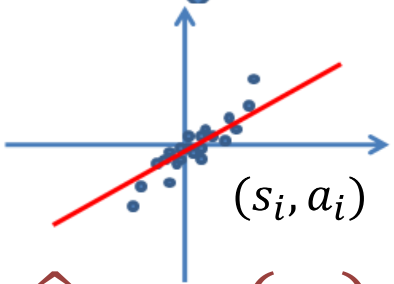
Expert Trajectories



s: state / observation

a: expert action

$$s_0, a_0, s_1, a_1, \dots, s_T, a_T$$



$$\hat{a} = \pi(s_i)$$

Supervised Learning

Learned Policy π

Mapping from state (image) to control (steering direction)

It Sounds Like Supervised Learning …

- High-dimensional observations:

[Pomerleau89, Saxena05, Ross11a]

Input:

Camera Image

Policy

Output:

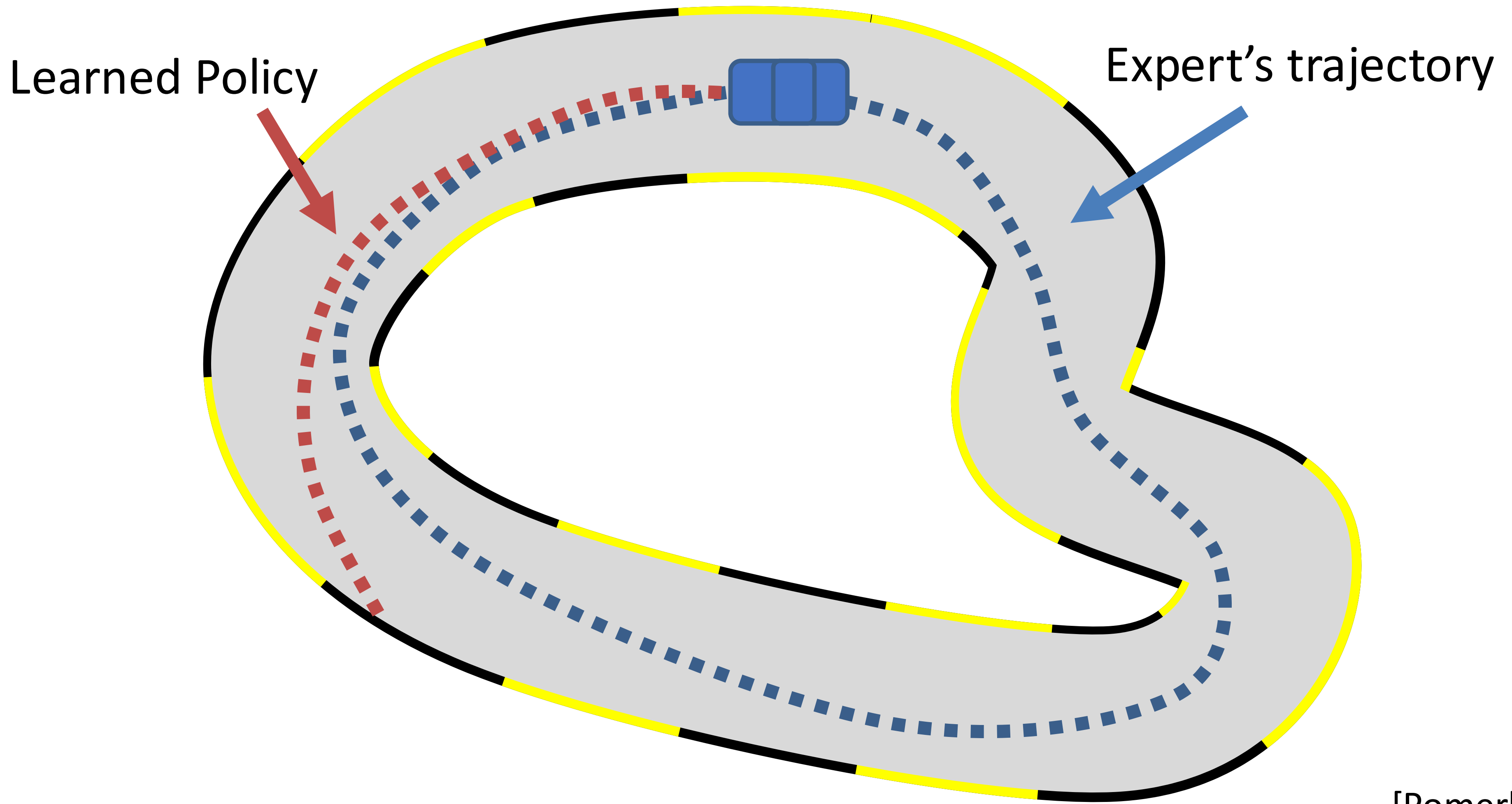
Steering Angle
in [-1, 1]

It Sounds Like Supervised Learning …

- ❑ Does BC work?

It Is NOT Standard Supervised Learning!

- Behavior cloning (BC) does work.
- What goes wrong?



[Pomerleau89,Daume09]

It Is NOT Standard Supervised Learning!

- ❑ The i.i.d. (independent and identically distributed) assumption doesn't hold!
- ❑ Recap the supervised learning story (lecture 3):

Theorem (Vapnik and Chervonenkis, 1971)

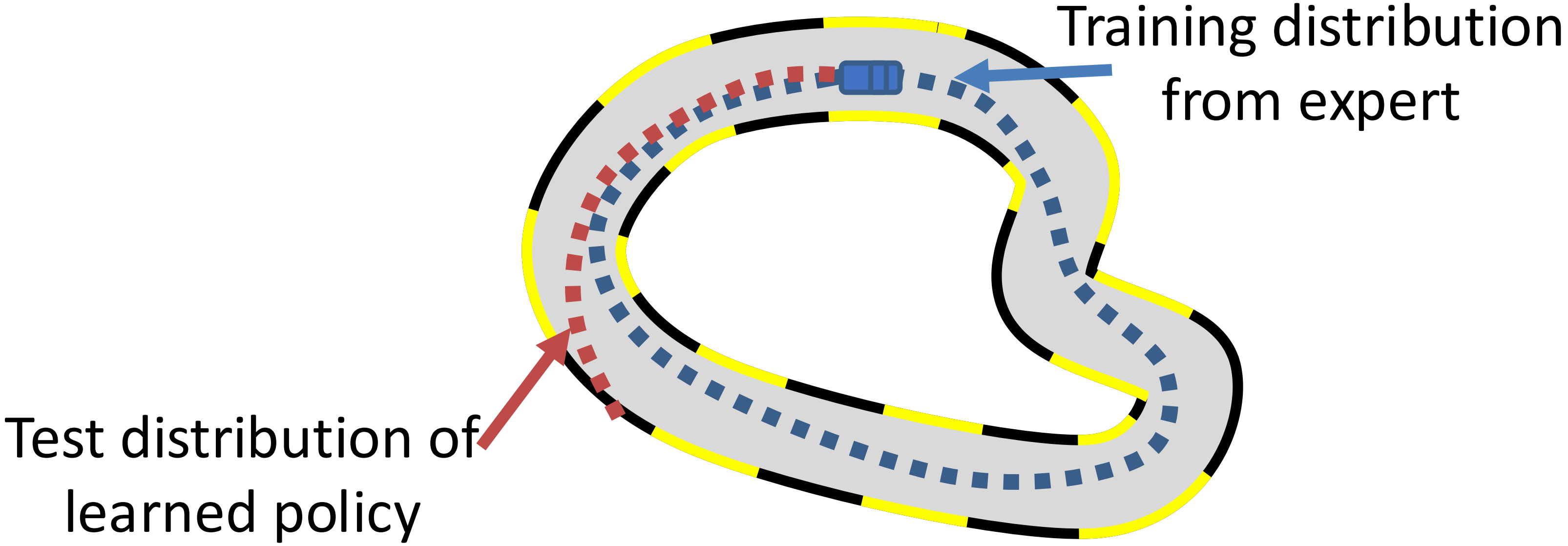
Consider $\mathcal{F} \subseteq \{-1,1\}^{\mathcal{X}}$, $\mathcal{Y} = \{\pm 1\}$, $\ell = \ell_{01}$.
For every prob distribution P on $\mathcal{X} \times \{-1,1\}$,
with probability $1-\delta$ over n iid examples $(x_1, y_1), \ldots, (x_n, y_n)$,
every f in $\mathcal{F}$ satisfies

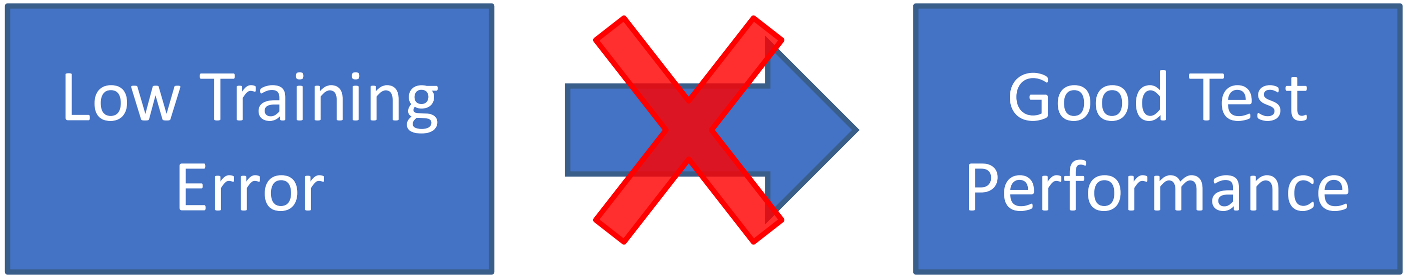
$$\overset{\text{“test error”}}{\mathbb{E}\ell_f} \le \overset{\text{“train error”}}{\hat{\mathbb{E}}\ell_f} + O\left(\sqrt{\frac{d_{VC}(\mathcal{F})}{n}}\right).$$

- ❑ (x_i, y_i) is i.i.d. sampled from some distribution
- ❑ Basically, “low training error” -> “good test performance”

It Is NOT Standard Supervised Learning!

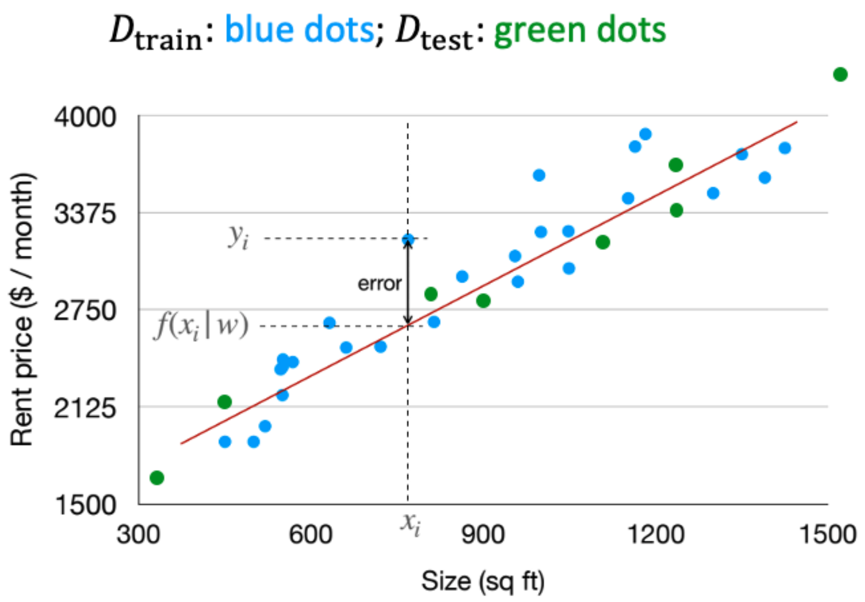
- ❑ The expert data only contains “good” behaviors
- ❑ Therefore, a small mistake in the test time will cause cascading failures

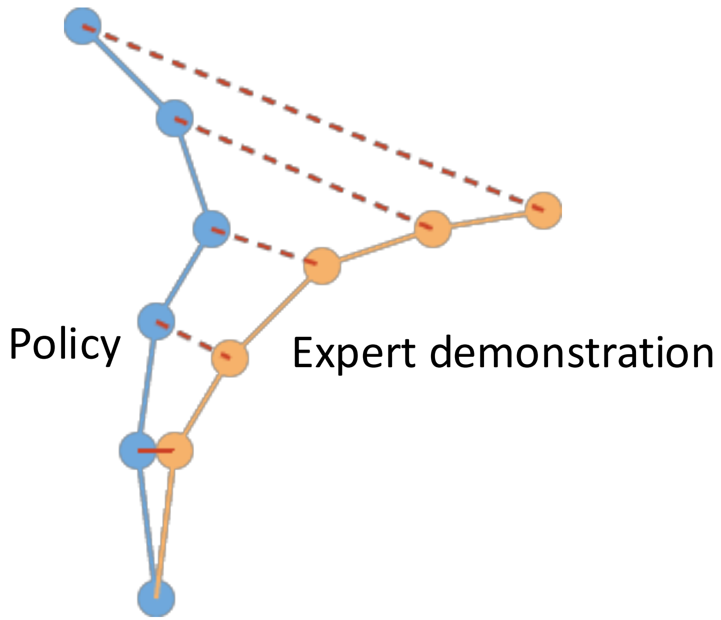




It Is NOT Standard Supervised Learning!

❑ Recap the **domain shift** problem in ML/DL (lecture 3):





❑ Even worse in sequential decision making! The domain shift is "**dynamic**" and "**fatal**"

- Policy only trained with expert data
- Expert has almost zero chance seeing "bad" state
- Statistically speaking, imitation learning is **cliff walking**
- Once we make one mistake, we are outside of the training distribution

Some Analysis

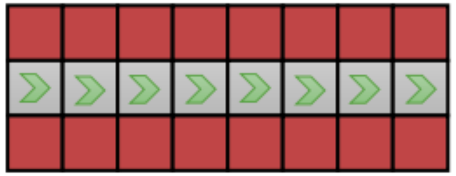
- ❑ Consider a "cliff walking" problem
  - Test error quadratic in # timesteps!

$$c(\mathbf{s},\mathbf{a}) = \begin{cases} 0 \text{ if } \mathbf{a} = \pi^\star(\mathbf{s}) \\ 1 \text{ otherwise} \end{cases}$$

assume: $\pi_\theta(\mathbf{a} \neq \pi^\star(\mathbf{s})|\mathbf{s}) \leq \epsilon$

for all $\mathbf{s} \in \mathcal{D}_{\text{train}}$

$$\underbrace{E\left[\sum_t c(\mathbf{s}_t,\mathbf{a}_t)\right]}_{O(\epsilon T^2)} \leq \epsilon T + \underbrace{\qquad\qquad\qquad}_{T \text{ terms, each } O(\epsilon T)}$$



- ❑ Why is it so pessimistic?
  - In reality, we can often **recover** from mistakes
  - But this **paradox** still exists: IL could work better if the data has more mistakes!

For more analysis, see Ross et al. "A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning"

This Issue is Also in Sequential Prediction

Ground Truth

Canonical Approach

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How to Address this Problem?

- ❑ Be smart about how we collect (and augment) our data
- ❑ Change the algorithm (DAgger)
- ❑ Next time: use very powerful models that make very few mistakes
- ❑ Lecture 23: use multi-task learning

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**ALVINN:
AN AUTONOMOUS LAND VEHICLE IN A
NEURAL NETWORK**

Dean A. Pomerleau
Computer Science Department
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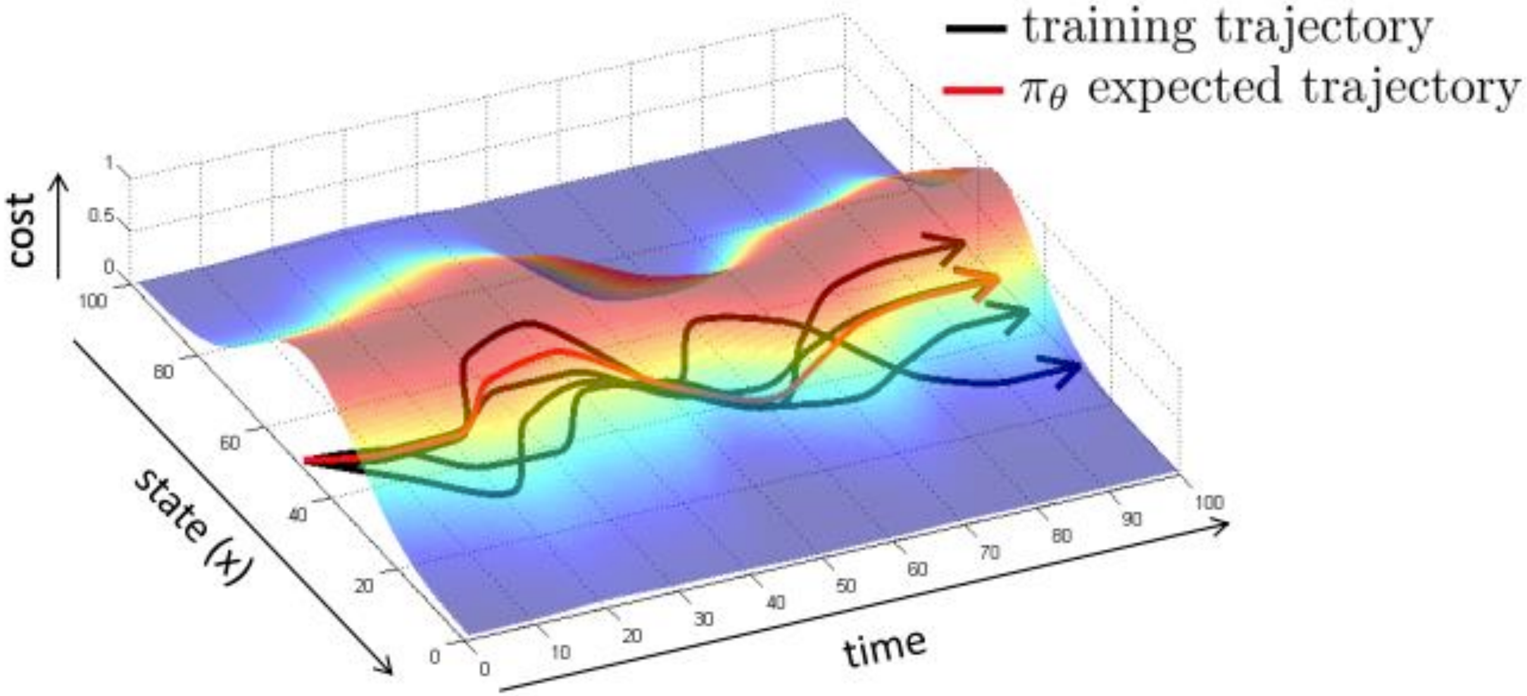
"...the network must not solely be shown examples of accurate driving, but also how to recover (i.e. return to the road center) once a mistake has been made."

What Makes BC Easy/Hard?

- ❑ Be smart about how we collect (and augment) our data
- ❑ Idea 1: Intentionally add mistakes and corrections
  - The mistakes hurt, but the corrections help, often more than the mistakes hurt
- ❑ Idea 2: Use data augmentation
  - Add some “fake” or “synthetic” data that illustrates corrections

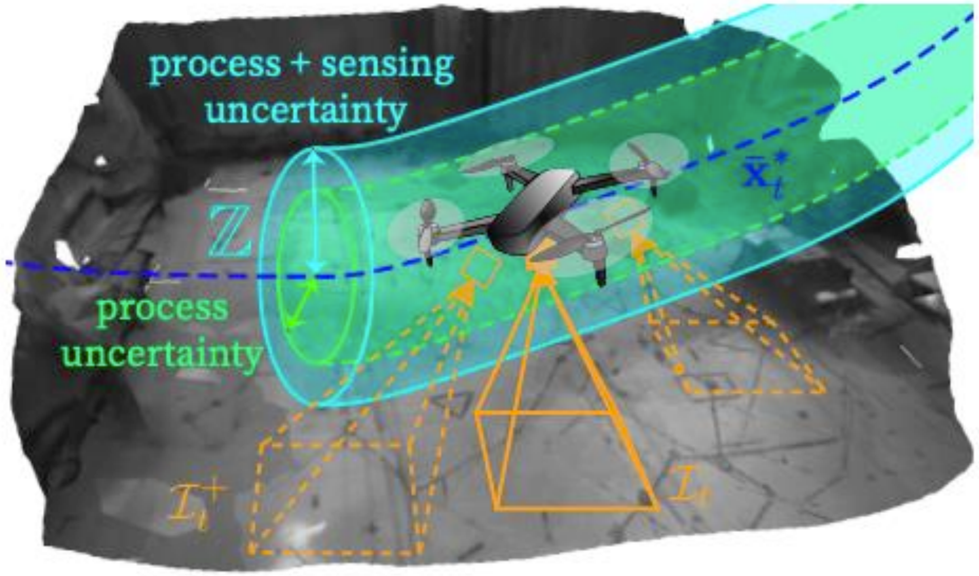


Data Augmentation Example

- ❑ Augment an MPC expert using a tube

Demonstration-Efficient Guided Policy Search via Imitation of Robust Tube MPC

Andrea Tagliabue, Dong-Ki Kim, Michael Everett, Jonathan P. How

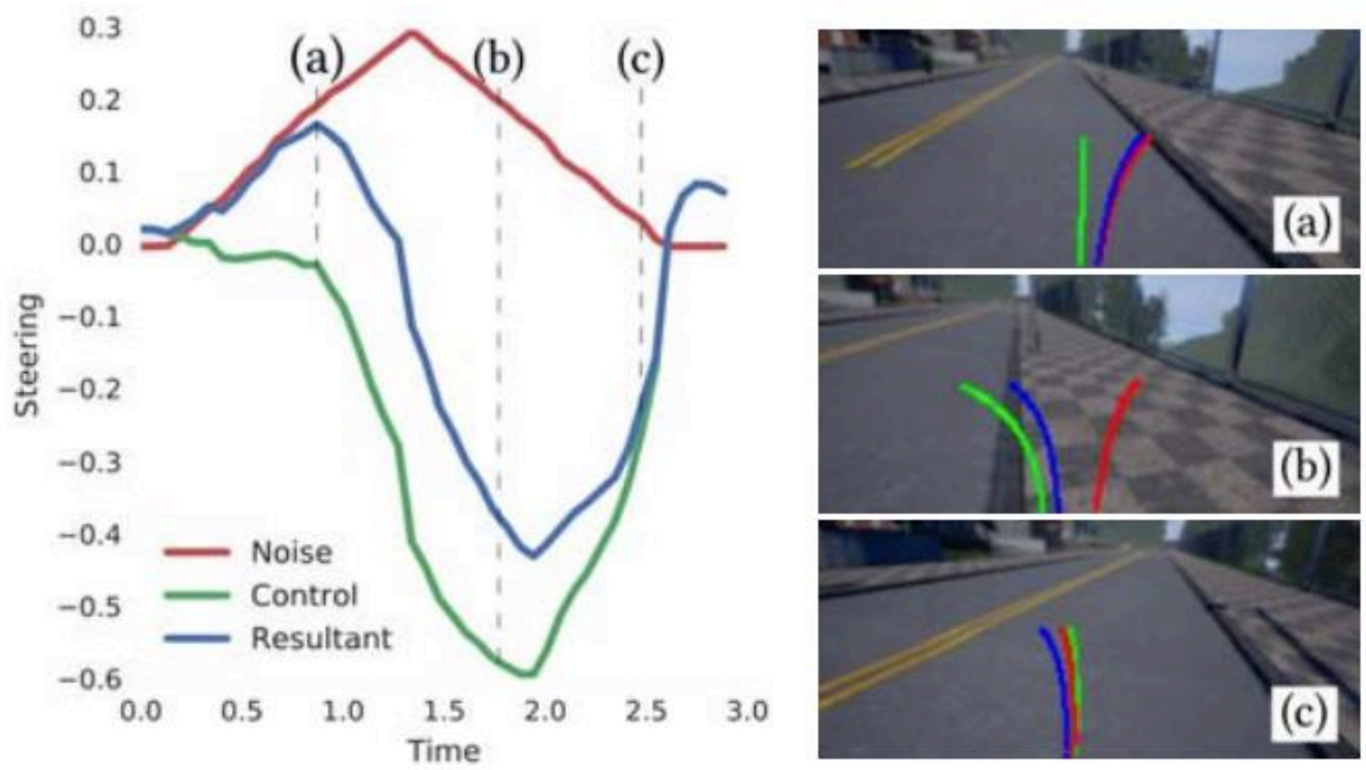


Data Augmentation Example

- ❑ Perturb the driver’s steering signal

End-to-end Driving via Conditional Imitation Learning

Felipe Codevilla[1,2] Matthias Müller[1,3] Antonio López[2] Vladlen Koltun[1] Alexey Dosovitskiy[1]



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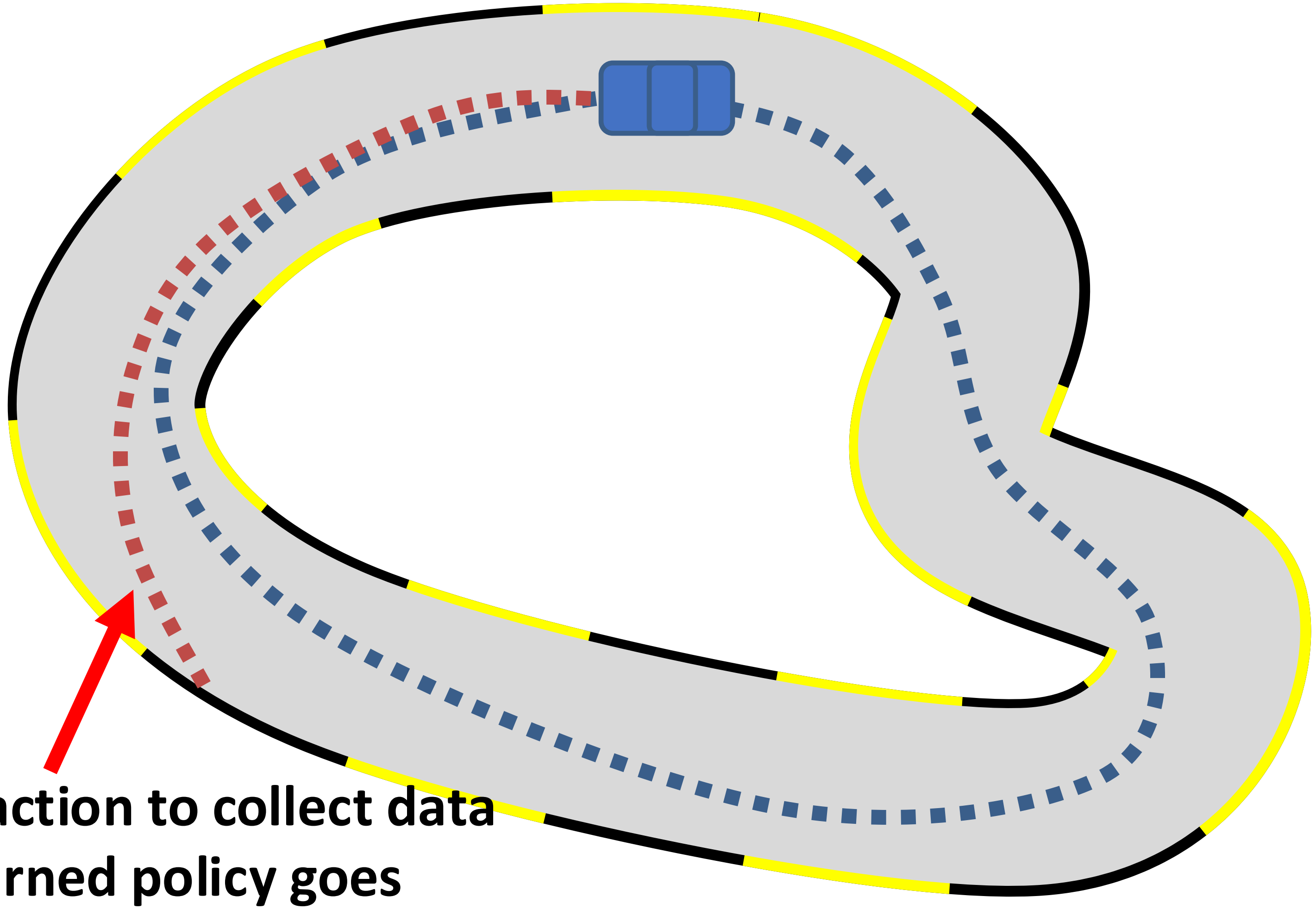
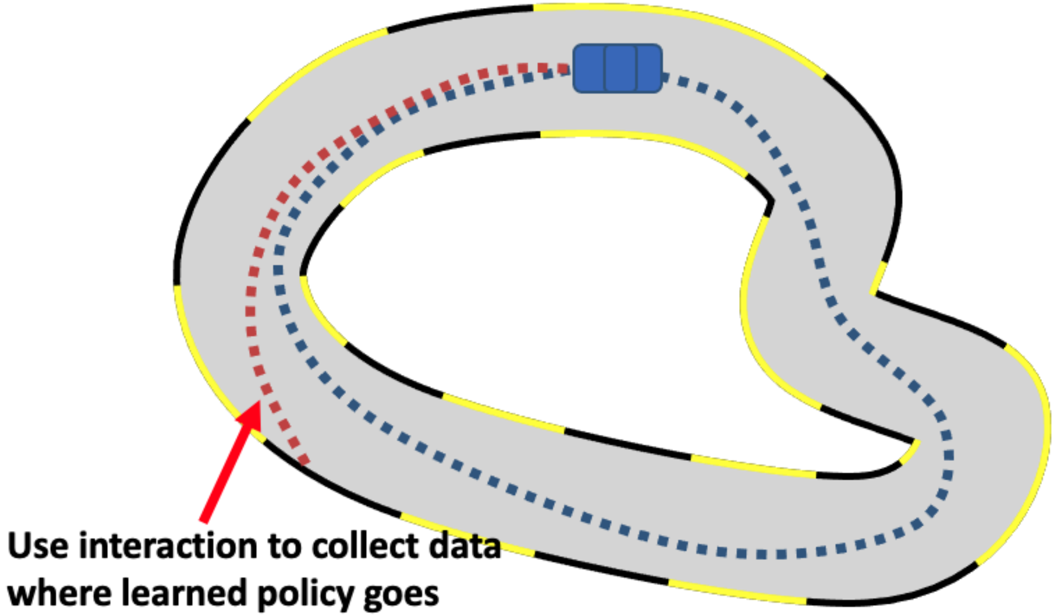
Key Idea: Interaction



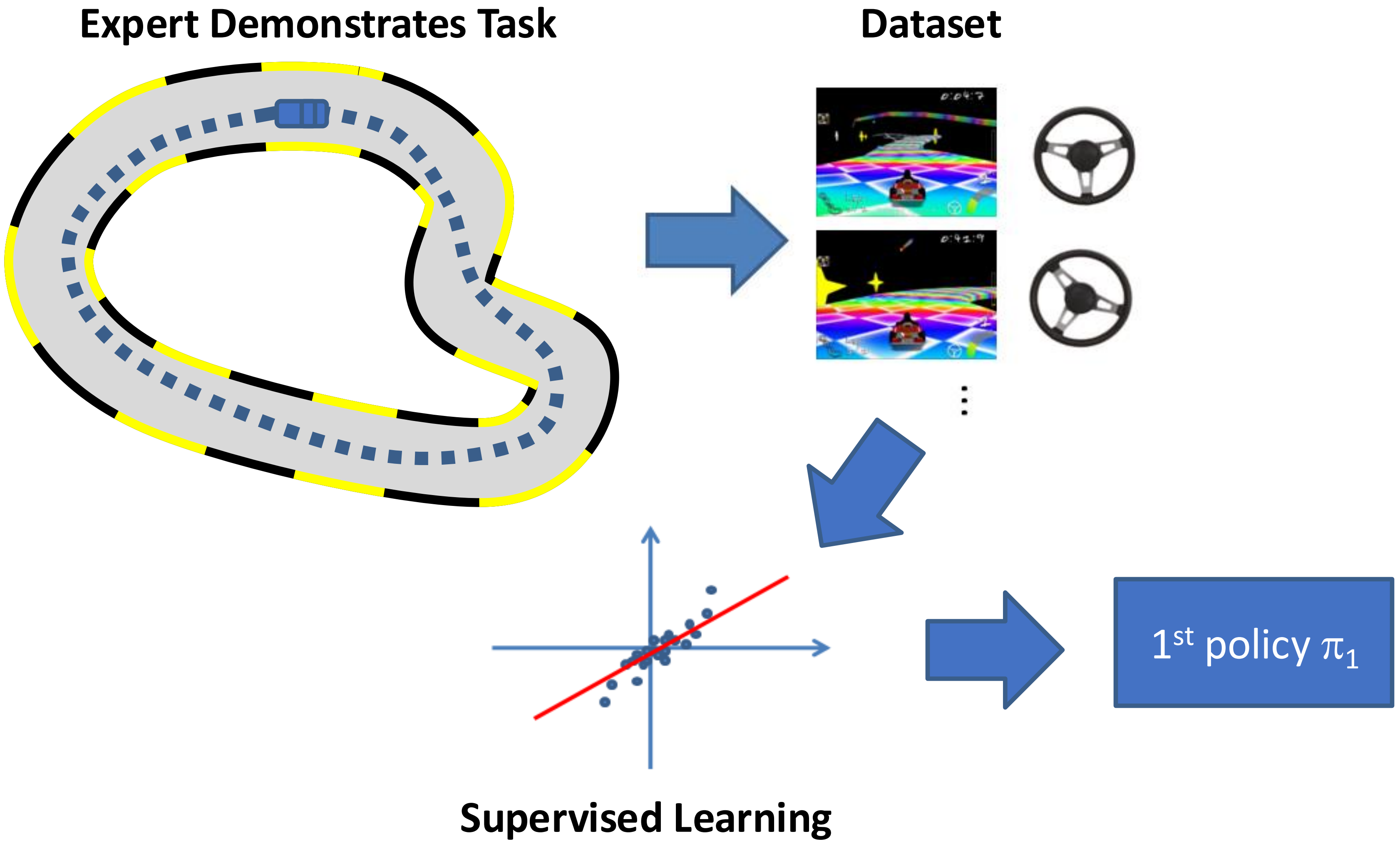
Use interaction to collect data where learned policy goes

Key Idea: Interaction



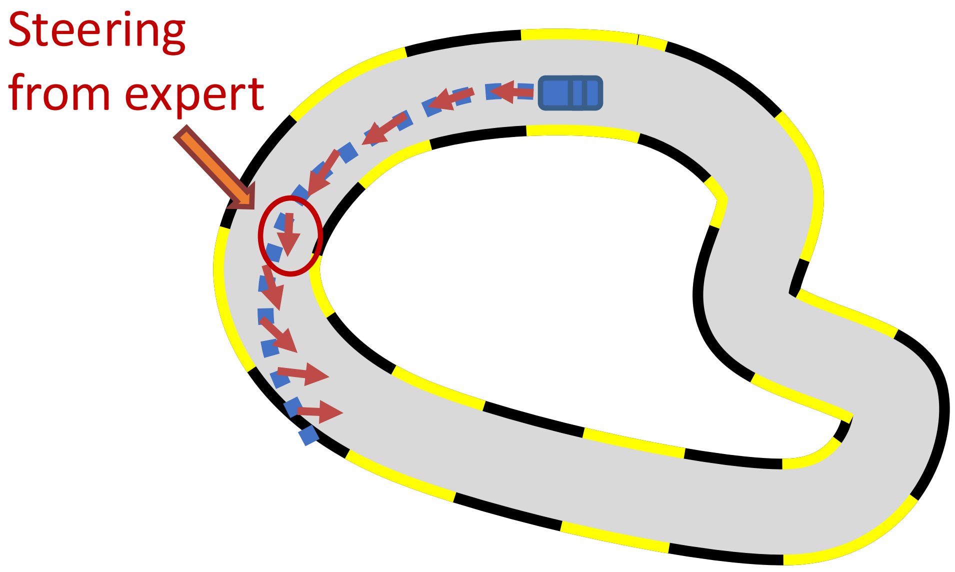
Train policy on expert data → Run policy → Collect new expert data to recover from mistakes

1^{st} iteration



2nd iteration

Execute π_1 and Query Expert



2^{nd} iteration

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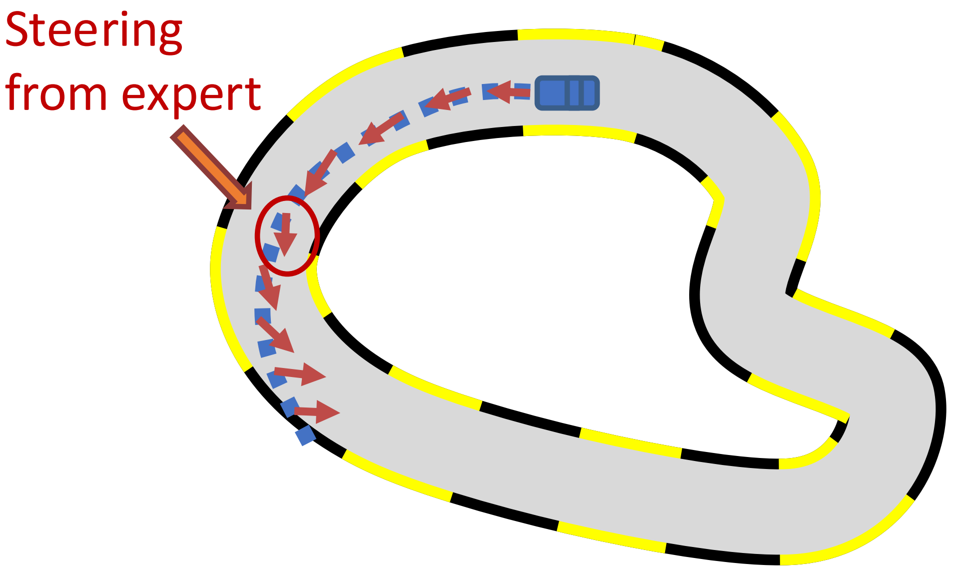
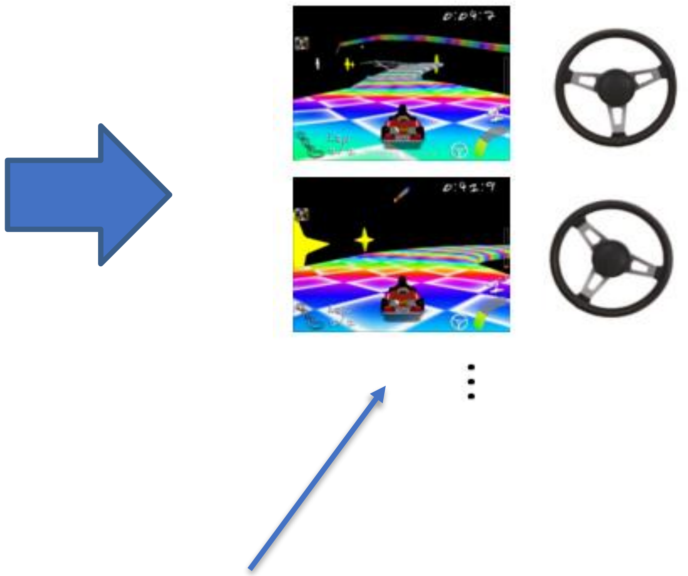


New Data

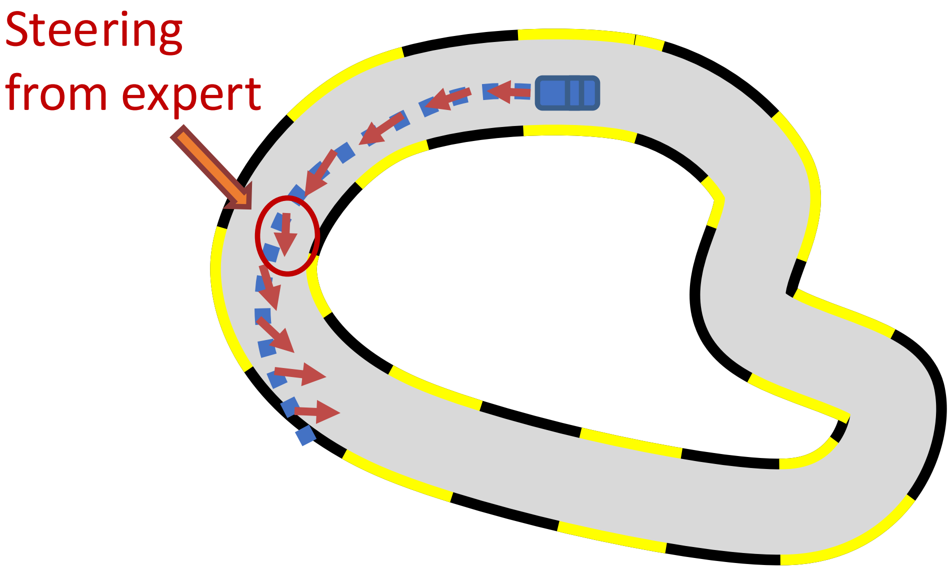


[Ross11a]

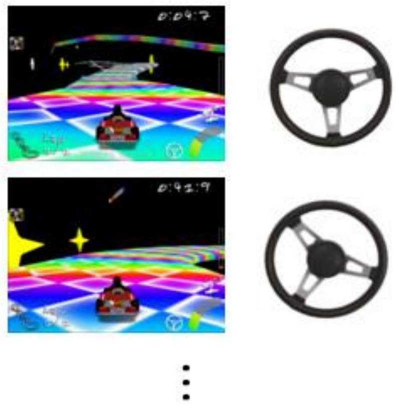
DAgger: Dataset Aggregation

2^{nd} iteration

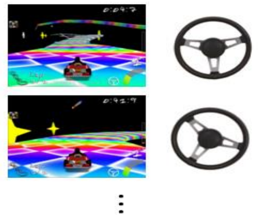
Execute π_1 and Query Expert



New Data



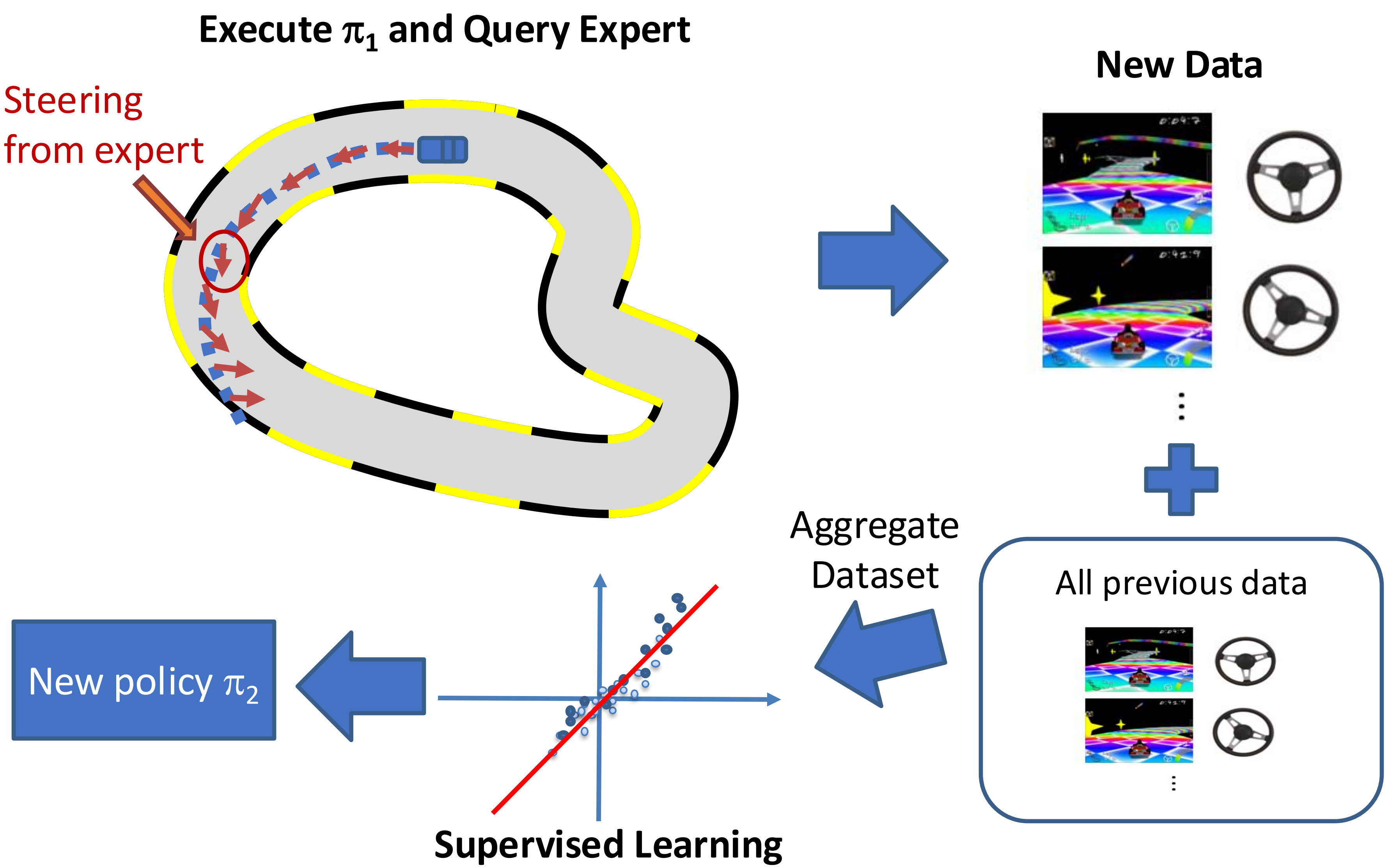
All previous data



[Ross11a]

DAgger: Dataset Aggregation

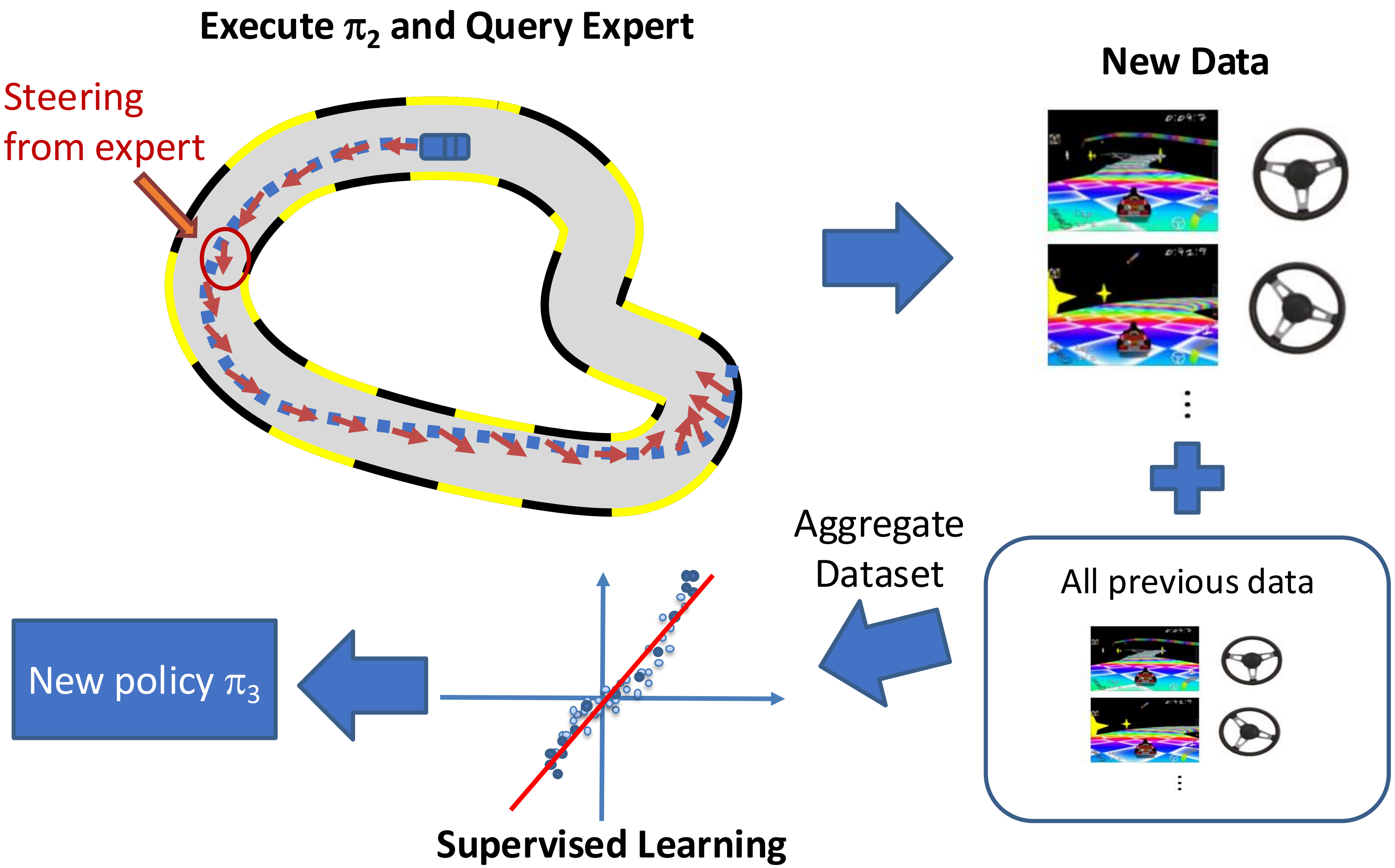
2^{nd} iteration



[Ross11a]

DAgger: Dataset Aggregation

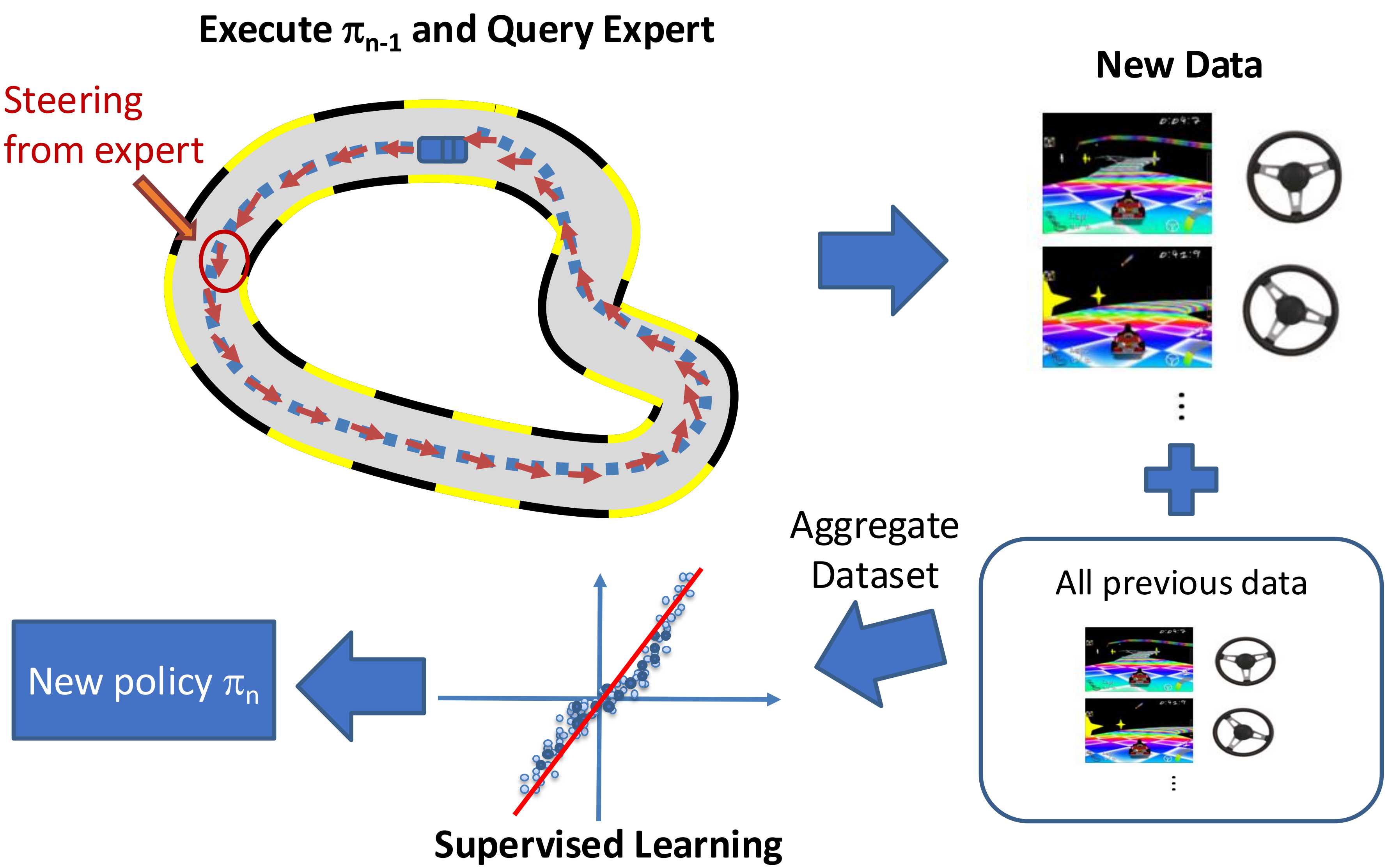
3^{rd} iteration



[Ross11a]

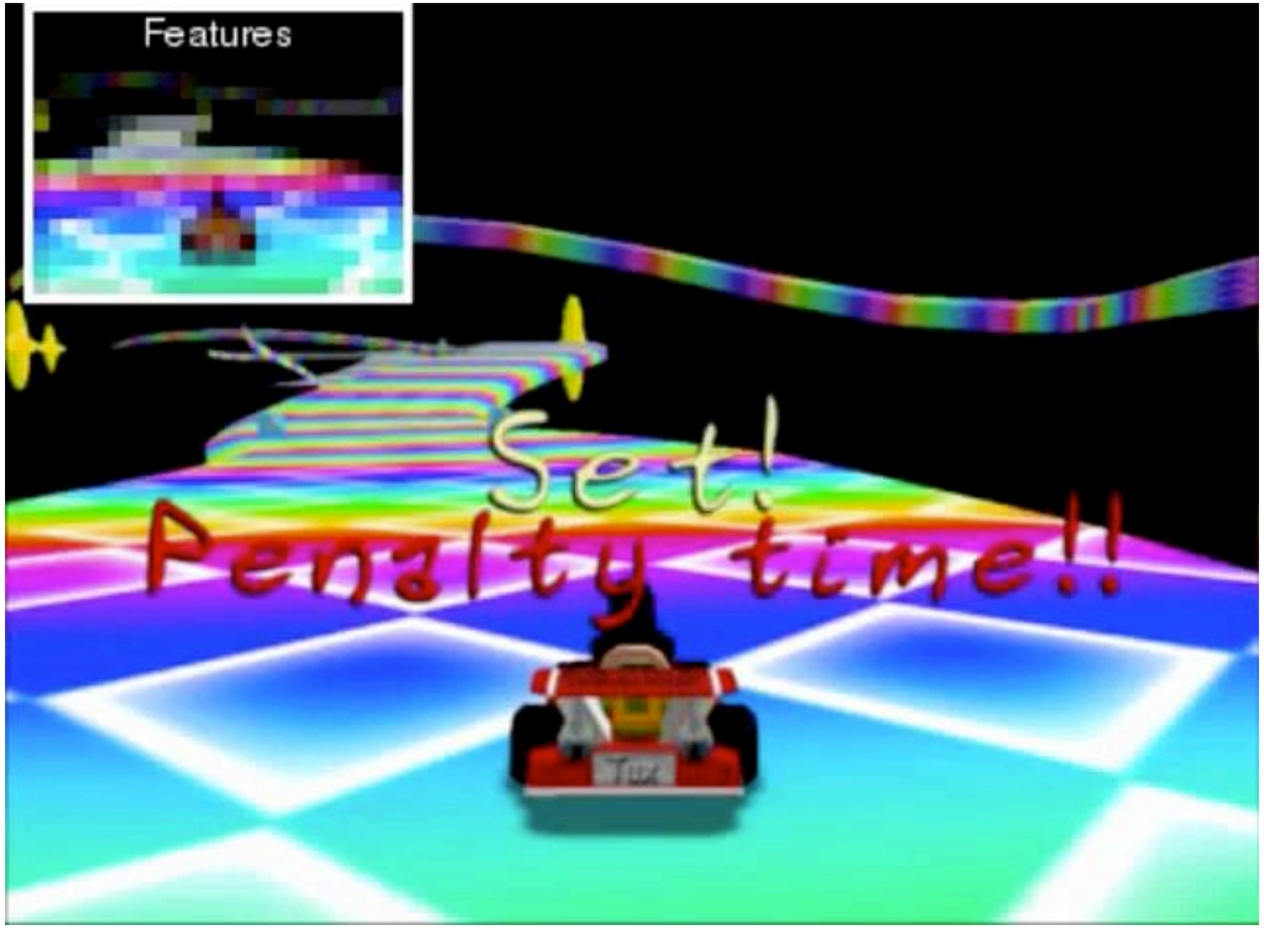
DAgger: Dataset Aggregation

n^{th} iteration



It Works!

- Left: BC; Right: DAgger



Summary of DAgger

- From CMU (2011)!
- It is essentially an online learning algorithm
  - With regret guarantees!
- Problem: need to keep querying the expert

1. train $\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$ from human data $\mathcal{D} = \{\mathbf{o}_1, \mathbf{a}_1, \ldots, \mathbf{o}_N, \mathbf{a}_N\}$
2. run $\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$ to get dataset $\mathcal{D}_\pi = \{\mathbf{o}_1, \ldots, \mathbf{o}_M\}$
3. Ask human to label $\mathcal{D}_\pi$ with actions $\mathbf{a}_t$
4. Aggregate: $\mathcal{D} \leftarrow \mathcal{D} \cup \mathcal{D}_\pi$

A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning

Stéphane Ross
Robotics Institute
Carnegie Mellon University
Pittsburgh, PA 15213, USA
stephaneross@cmu.edu

Geoffrey J. Gordon
Machine Learning Department
Carnegie Mellon University
Pittsburgh, PA 15213, USA
ggordon@cs.cmu.edu

J. Andrew Bagnell
Robotics Institute
Carnegie Mellon University
Pittsburgh, PA 15213, USA
dbagnell@ri.cmu.edu